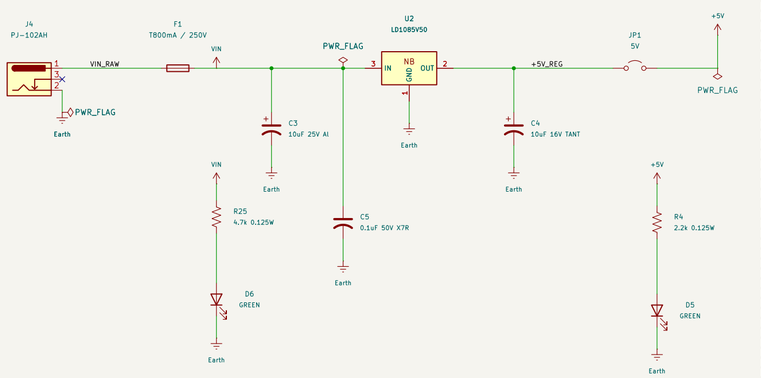
Fill-A-Bot Main Control Board

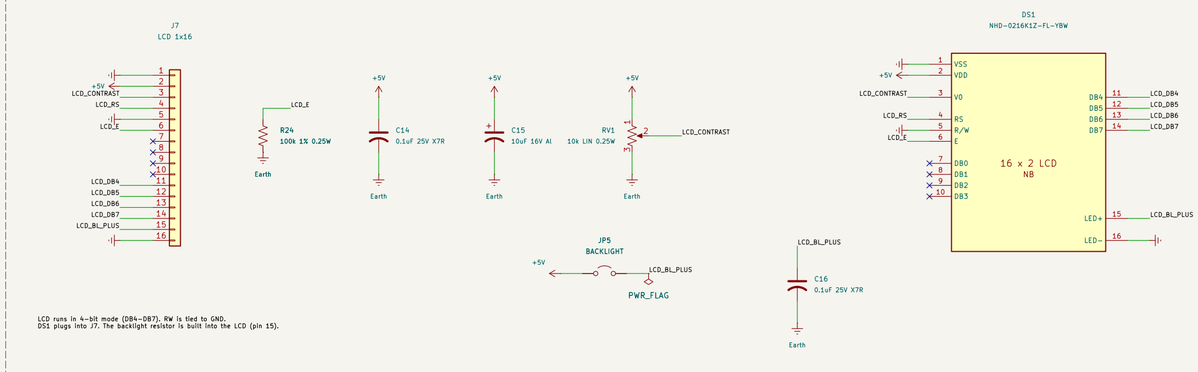
Power Input and 5V Regulator



F1 is a 5x20mm fuse in a holder so it can be swapped.

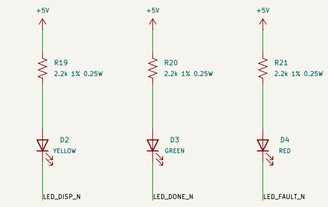
Input is 9–12V DC, center positive.
U2 gets a heat sink (HSE14-38+22SP).
D5 and D6 light up when the 5V and 9V rails are on.

16x2 LCD

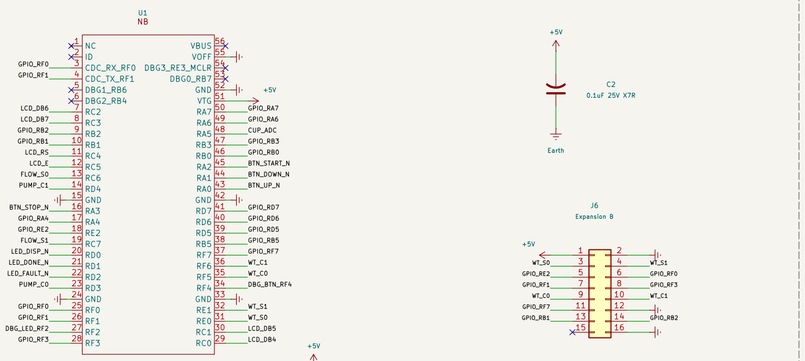


LCD runs in 4-bit mode (DB4–DB7), RW is tied to GND.
DS1 plugs into J7. The backlight resistor is built into the LCD (pin 15).

Status LEDs

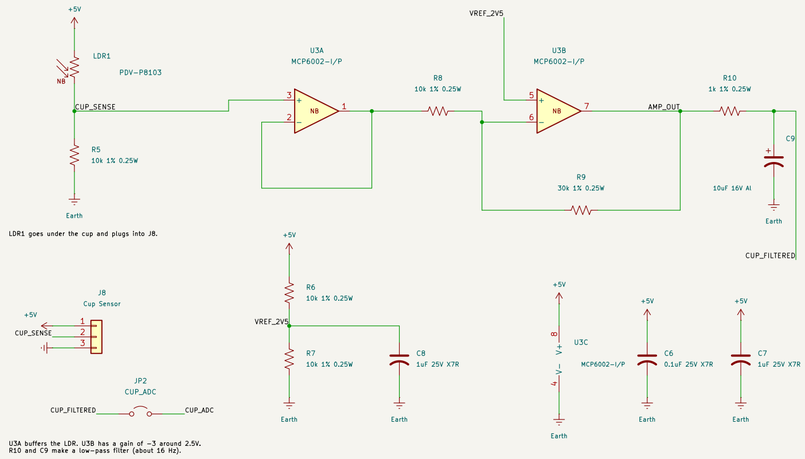


Curiosity Nano, Extra Headers, Debug



Extra headers for spare pins

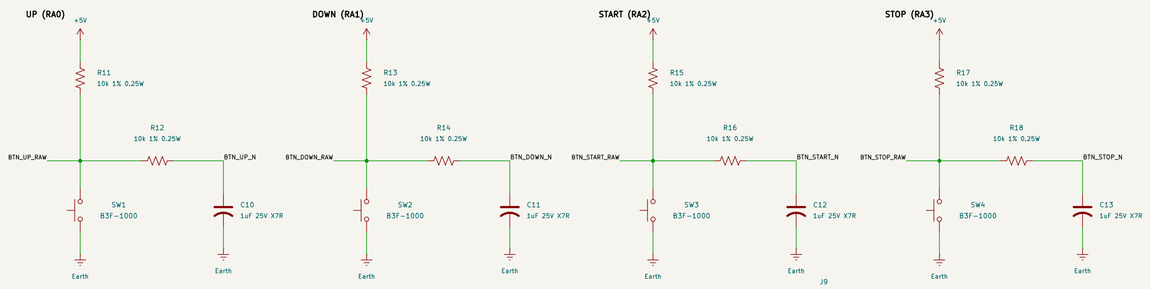
Cup Sensor (LDR + Op Amp)



LDR1 goes under the cup and plugs into J8.

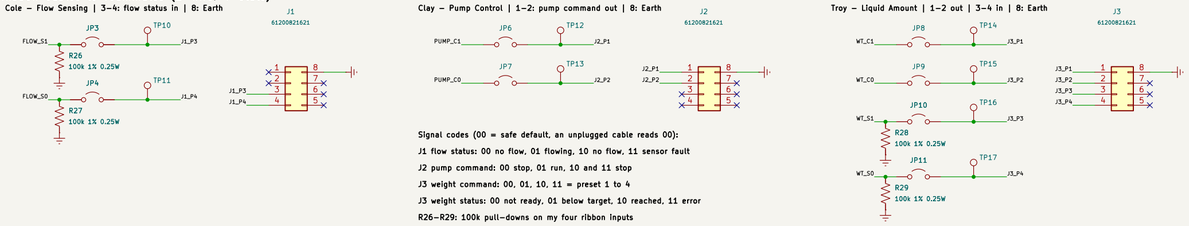
U3A buffers the LDR. U3B has a gain of −3 around 2.5V.
R10 and C9 make a low-pass filter (about 16 Hz).

Buttons



Each button has a 10k pull-up and a 10k/1uF RC debounce.
J9 is for buttons mounted off the board.

Ribbon Cables to Teammates (2-wire, 4-state)



Signal codes (00 = safe default, an unplugged cable reads 00):
J1 flow status: 00 no flow, 01 flowing, 10 no flow, 11 sensor fault
J2 pump command: 00 stop, 01 run, 10 and 11 stop
J3 weight command: 00, 01, 10, 11 = preset 1 to 4
J3 weight status: 00 not ready, 01 below target, 10 reached, 11 error
R26–R29: 100k pull-downs on my four ribbon inputs

Nikita Bangiyev – EGR 304 Team 201

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File: EGR304_Main_Control_Nikita.kicad_sch

Title: Fill-A-Bot Main Control Board

Size: A2	Date: 2026-11-04	Rev: 5
KiCad E.D.A. 10.0.8		Id: 1/1